The psychology of language – E578

The semantic blocking effect

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– Presentation of the task and effect
– A first explanation
– Problematic results
– Alternative interpretation

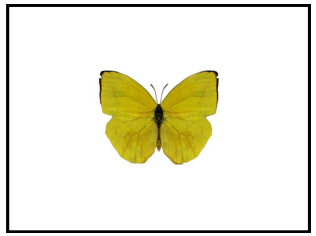
Experimental design

Naming task: An image is given, and people are asked to name it.
Ⓟ We measure response times. Usually recorded using an automated key.

Experimental design

One cycle

Heterogeneous set



« butterfly »

« cauliflower »

« fork »

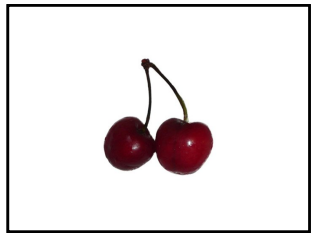
« glove »

Time

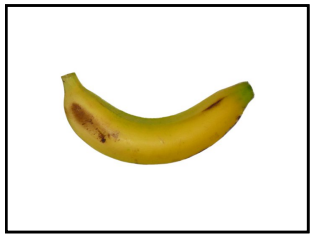
Experimental design

One cycle

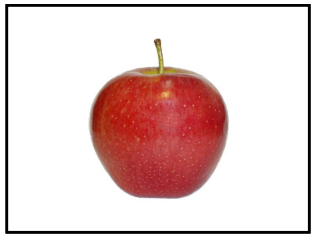
Homogeneous set



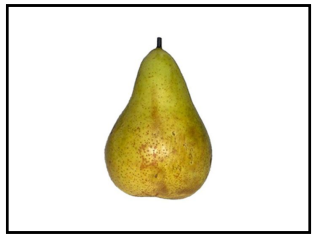
« cherry »



« banana »



« apple »



« pear »

Time

Experimental design

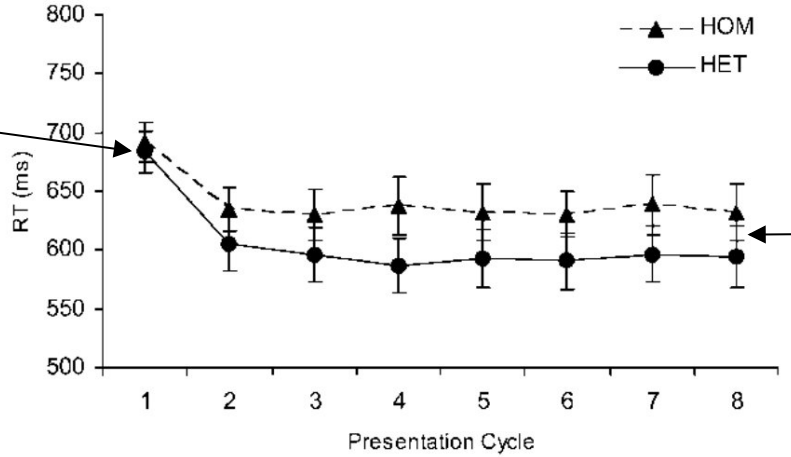
Cycle 1 – Cycle 2 – Cycle 3 – Cycle 4…

Same objects, different presentation order

We are interested to **compare** homogeneous and heterogeneous sets

Experimental design

No effect in the first cycle



Slower naming latencies for homogeneous sets from cycle 2 on
Ⓟ **The refractory effect**.

Figure 2. Experiment 1: Mean response times (RT) by context (homogeneous, hom vs. heterogeneous, het) and presentation cycle. The error bars represent one standard error.

Belke, Meyer, and Damian (2005)

What’s happening *inside* a cycle?

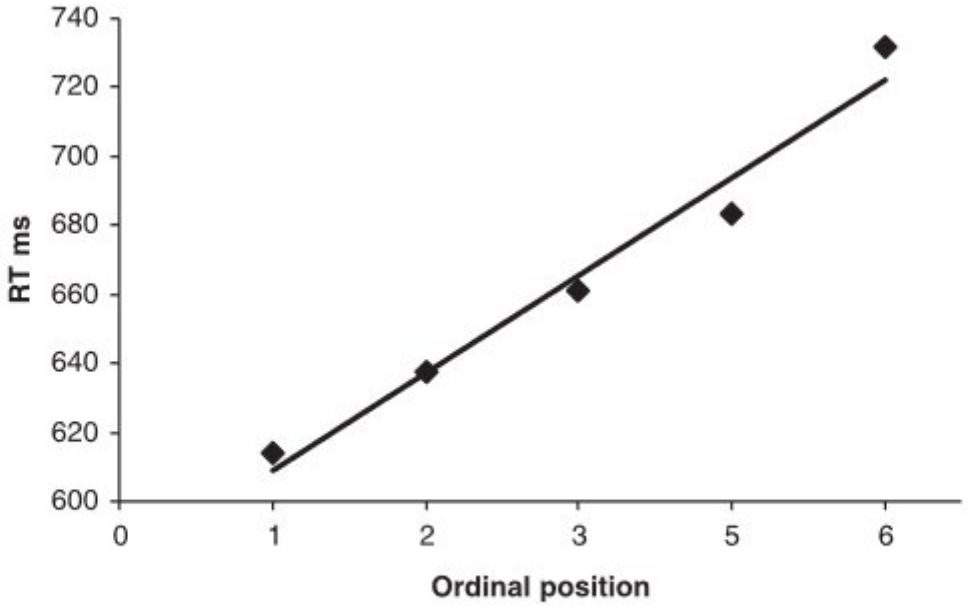


Fig. 1. The effect of ordinal position within a category on naming reaction time (uncorrected RTs).

Howard et al. (2006)

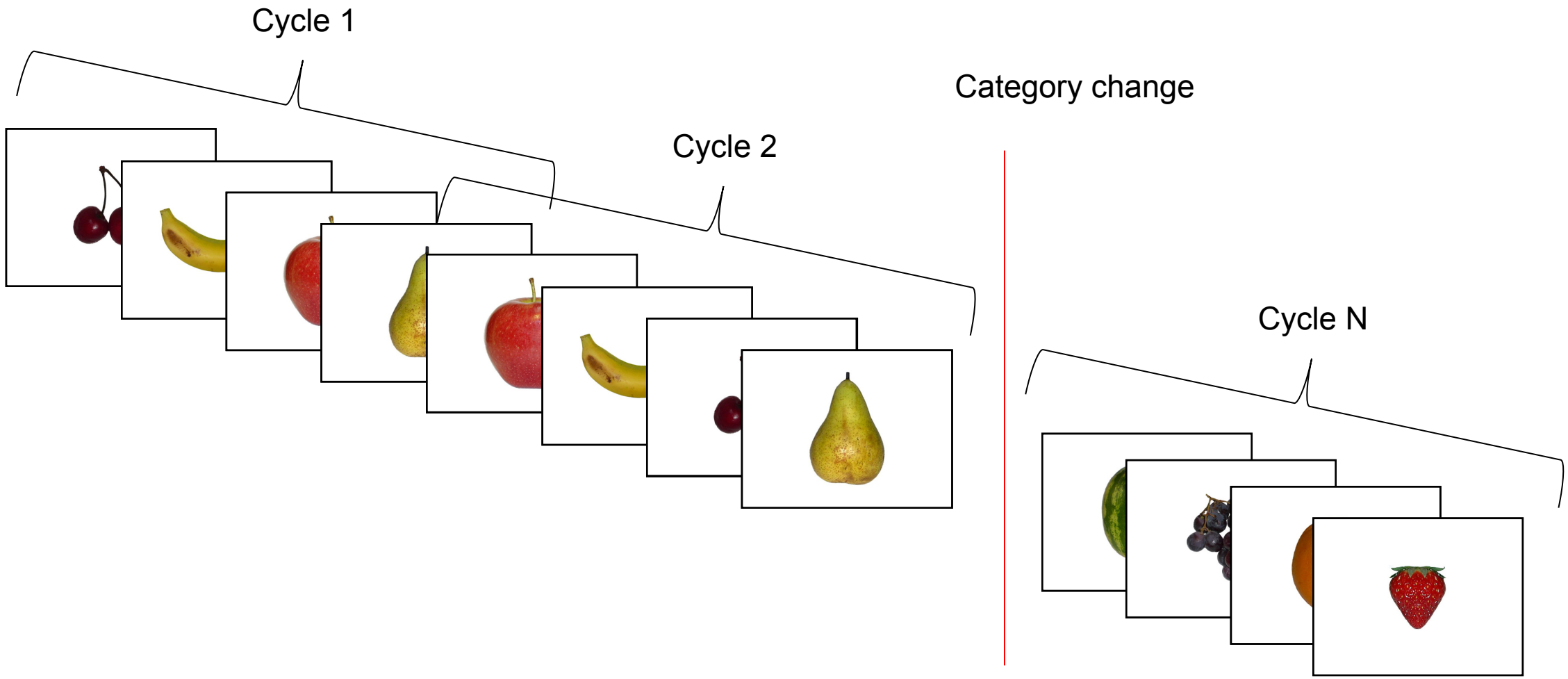
Experimental design

In this experiment, all stimuli were kept identical throughout successive presentation cycles.

Is the effect specific to the repeated items?

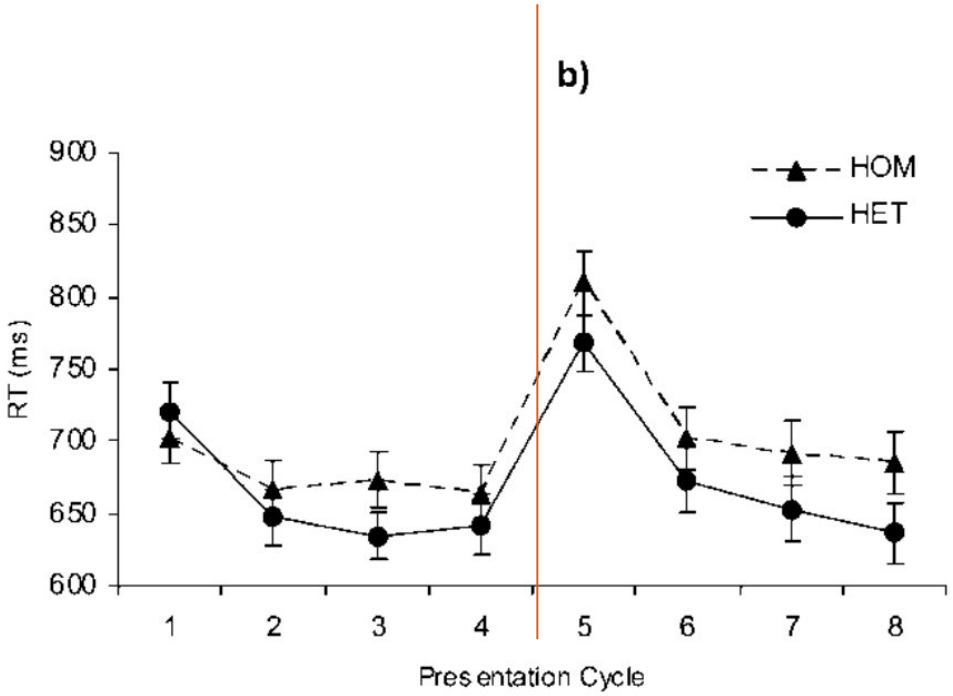
Solution: we can test whether it will generalize for not-yet-presented items from the same category.

Belke, Meyer, and Damian (2005)



Belke, Meyer, and Damian (2005)

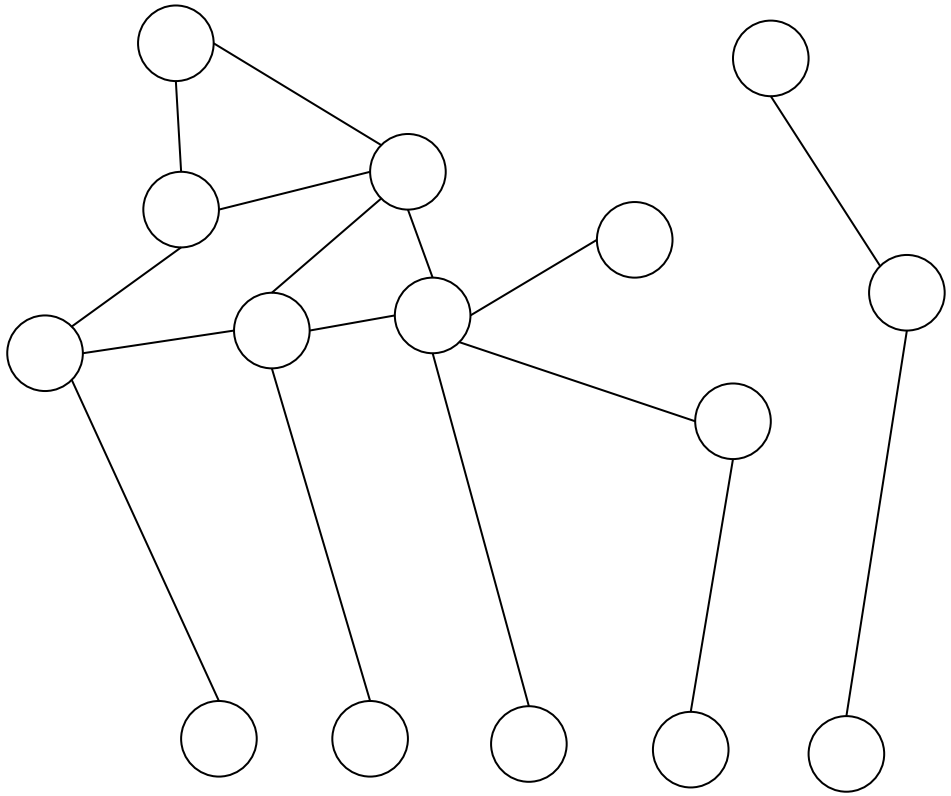
Category change



The refractory effect expands to new items!

Belke, Meyer, and Damian (2005)

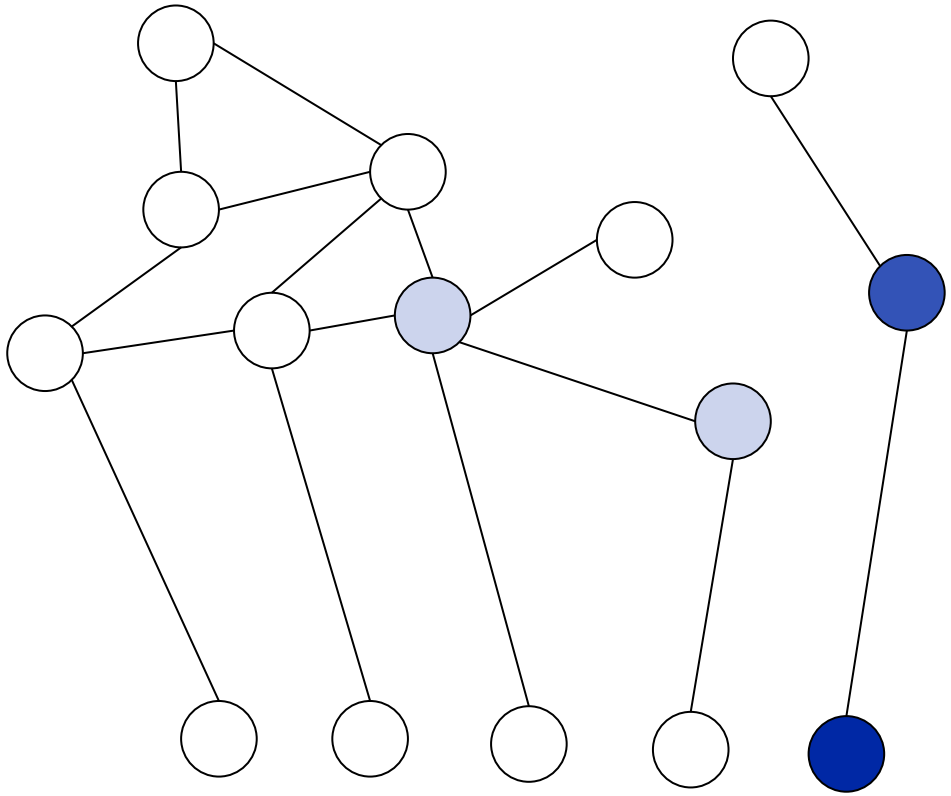
How can we explain the semantic blocking effect?
What does it tell us about the language production system?



Semantic system

Lexical system

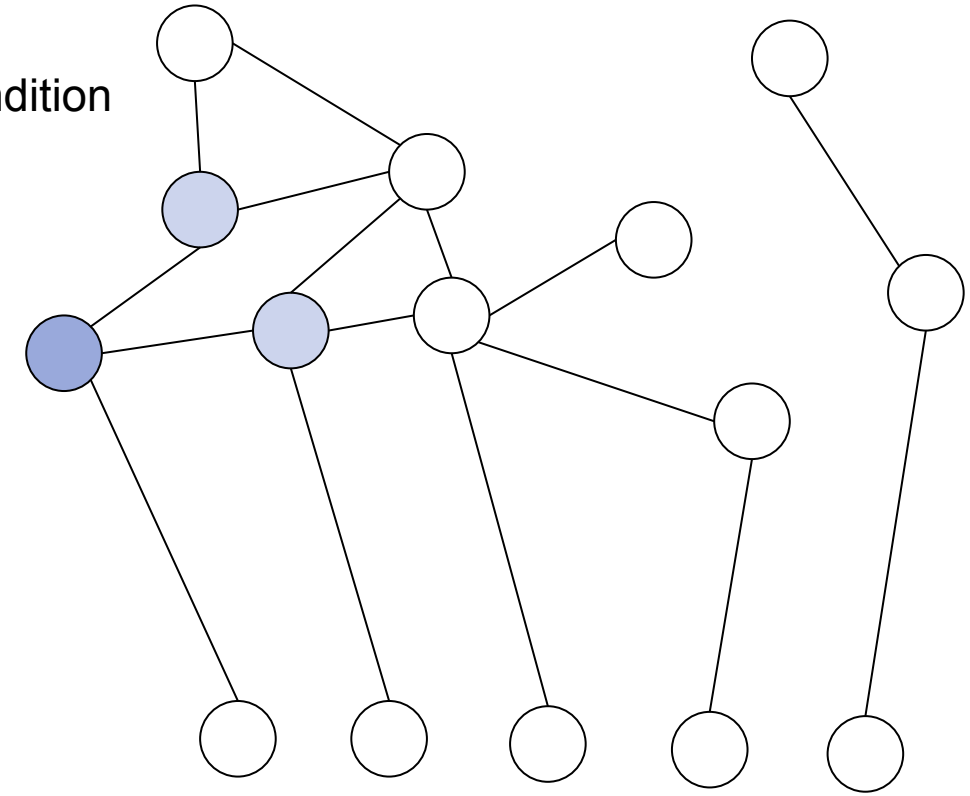
Heterogeneous condition



Semantic system

Lexical system

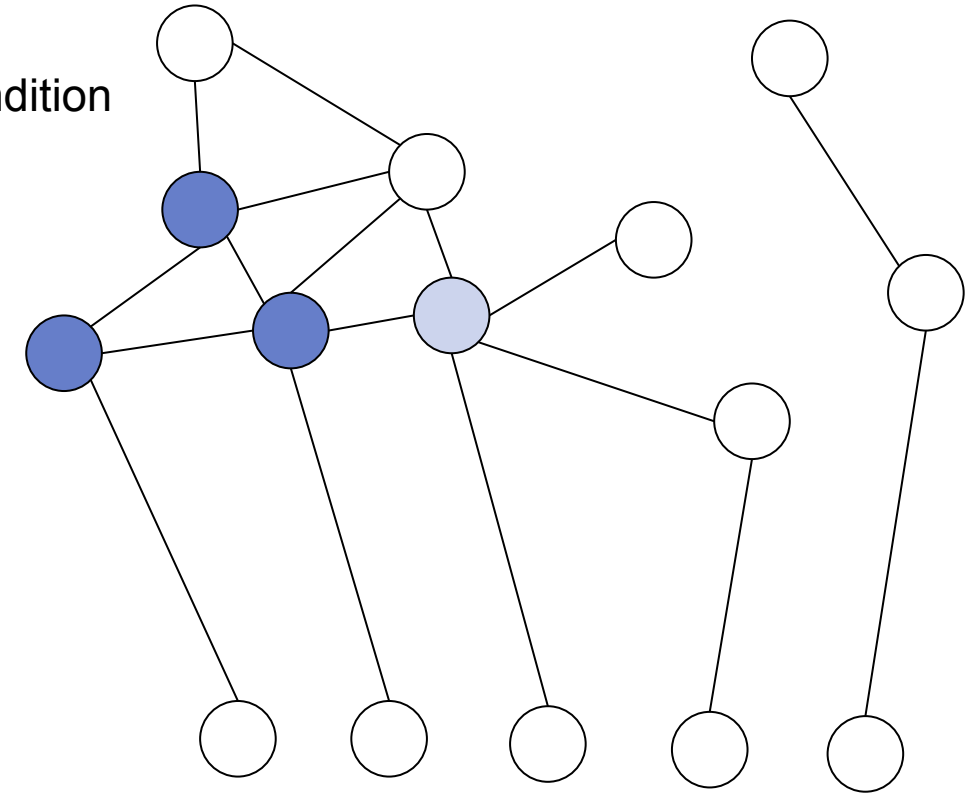
Homogeneous condition



Semantic system

Lexical system

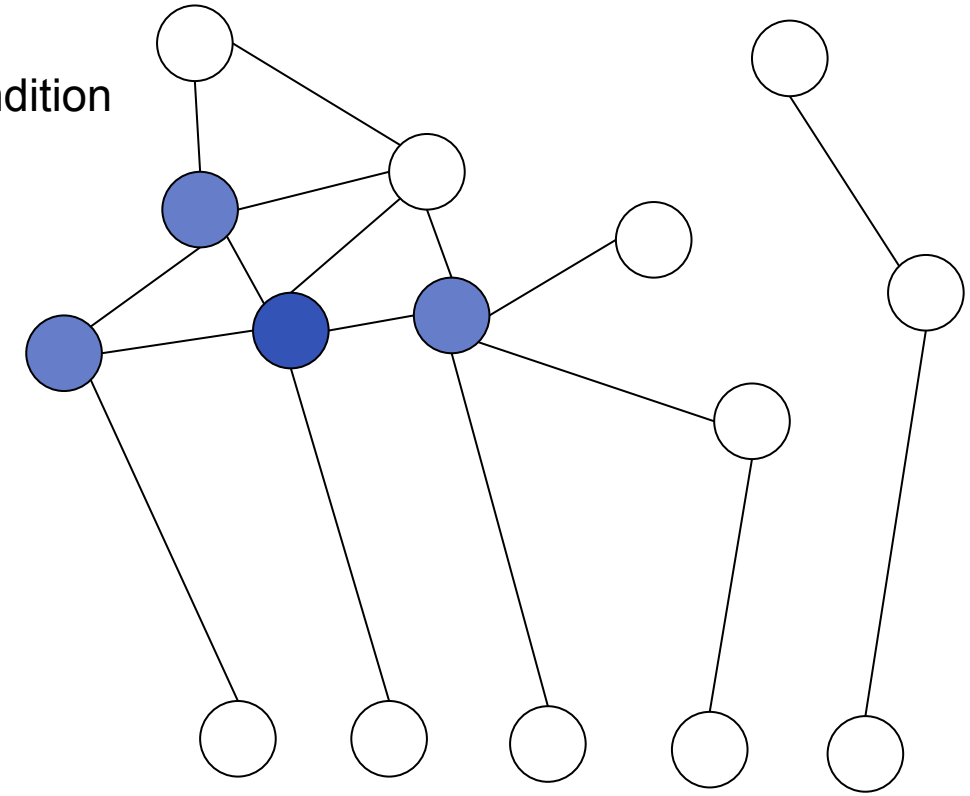
Homogeneous condition



Semantic system

Lexical system

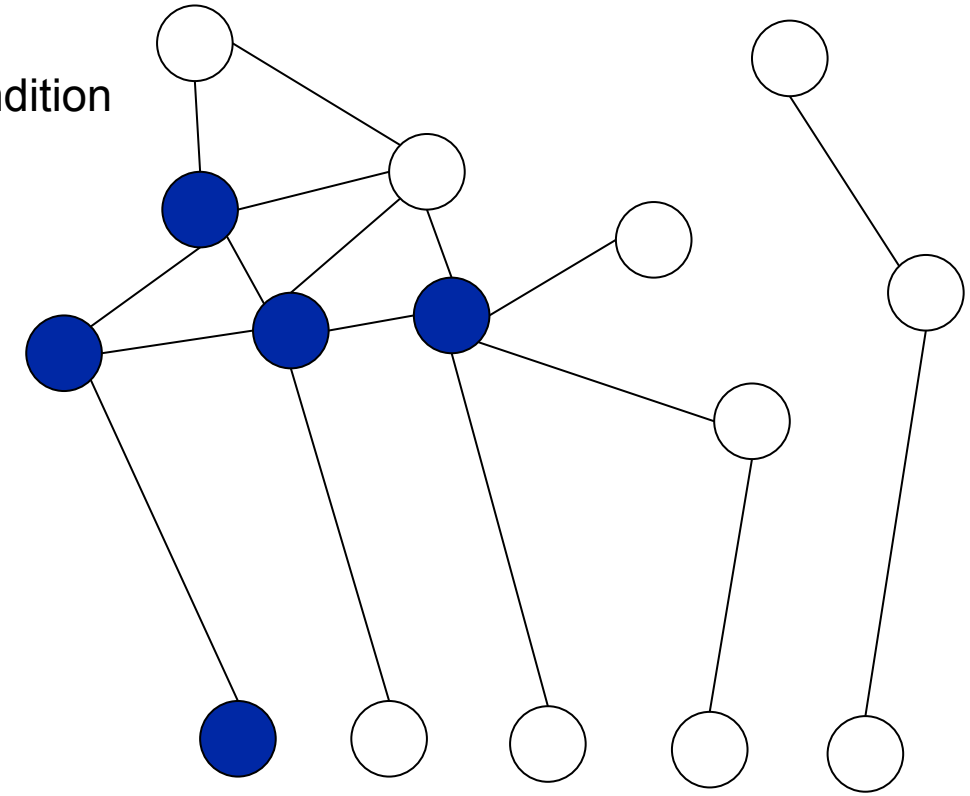
Homogeneous condition



Semantic system

Lexical system

Homogeneous condition



Semantic system

Lexical system

Luce's choice rule:

$$P = \frac{A}{\sum_{k=1}^{N} A}$$

Heterogeneous condition A (Activation) = [0.90; 0.30; 0.10; 0.10]

0.9 / sum(A) 0.3 / sum(A) 0.1 / sum(A) 0.1 / sum(A)

The sum of this vector is equal to one ℗ P = [**0.64**; 0.21; 0.07; 0.07]

These values represent the **competition** between items. Here, the **target** is associated to a large value relatively to other competitors, so there's not much competition.

Luce's choice rule:

$$P = \frac{A}{\sum_{k=1}^{N} A}$$

Homogeneous condition A (Activation) = [0.95; 0.80; 0.91; 0.90]

0.95 / sum(A) 0.8 / sum(A) 0.91 / sum(A) 0.90 / sum(A)

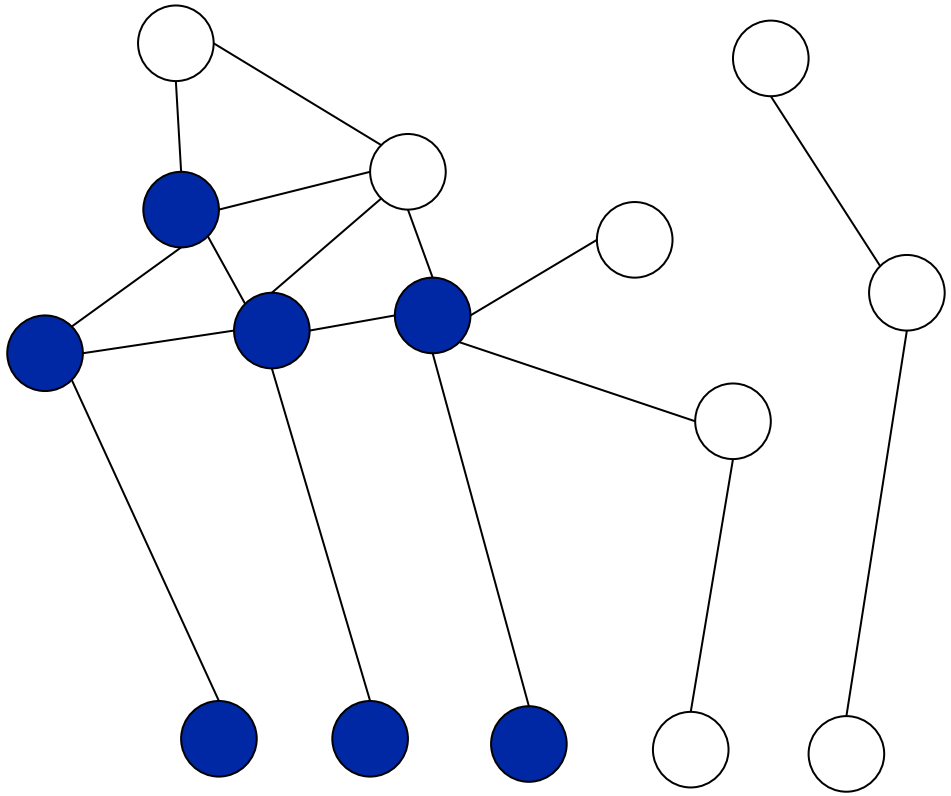
P = [**0.27**; 0.22; 0.26; 0.25]

Increased competition!

Explained verbally:

Items spread activation in a semantic network.
The homogenenous condition creates an over-activation in this network.
Since selection is activation-based, an over-activation of semantically similar competitors creates a higher competition between candidates.

The Luce's choice rule is simply a mathematical expression of this idea.

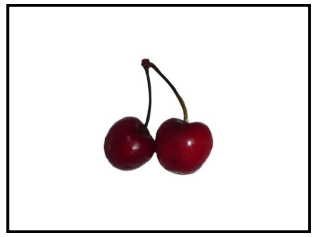


Semantic system ?

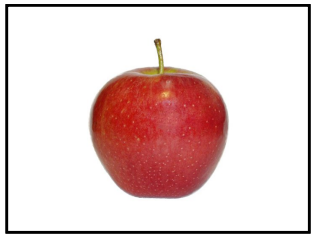
Lexical system ?

Reading words aloud relies less strongly on the semantic system than naming objects

But both rely on the selection of a lexical candidate that must be output



banana



pear

Time

Kroll & Stewart (1994)

If the competition occurs at the lexical stage, then the blocking effect should be preserved in these conditions.
If not, we can infer that the competition must occurs in the semantic system.

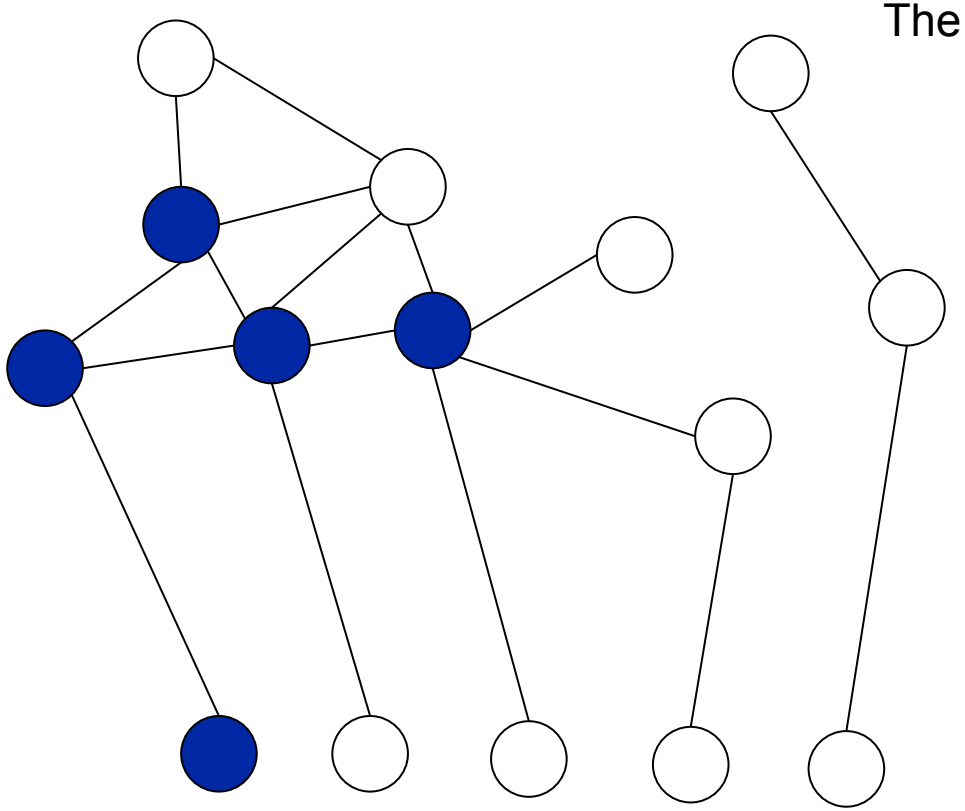
Kroll & Stewart (1994)

TABLE 1
MEAN RESPONSE LATENCIES (IN MILLISECONDS) AND PERCENTAGE ERRORS (AS SHOWN IN PARENTHESES) TO NAME WORDS AND PICTURES IN EXPERIMENTS 1 AND 2 IN CATEGORIZED AND RANDOMIZED LIST CONTEXTS AND MEAN PERCENTAGE INCIDENTAL RECALL IN EXPERIMENT 1

List Condition	Target modality: Words	Target modality: Pictures
Experiment 1: Blocked modality		
Categorized lists	514 (0.0%)	819 (7.3%)
Mean % recall	18.5%	43.2%
Randomized lists	516 (0.0%)	783 (6.4%)
Mean % recall	13.3%	27.4%
Magnitude of category interference	−2 ms	36 ms
Experiment 2: Alternating modalities		
Categorized lists	549 (2.1%)	792 (14.7%)
Randomized lists	542 (2.3%)	798 (11.4%)
Magnitude of category interference	7 ms	−6 ms

Alternating words and pictures makes the blocking effect disappear!

Kroll & Stewart (1994)



The blocking effect is due to a difficulty to select concepts in semantic memory

Semantic system

Lexical system

Is there a possibility to test this **priming** idea?

One characteristic of priming is that it is **short-lived**. It doesn't last over time.

Let's manipulate time in the blocking effect.

Naming task interleaved by mental rotations (i.e., visuospatial distractors)

« cherry » « banana » « apple » « pear »

We can manipulate the number of distractors (e.g., 0; 1; 2)

Time

Damian & Als (2005)

Table 1
Experiment 1: Response Latency Means and Standard Deviations of Picture-Naming Responses by Number of Intervening Mental Rotation Trials (0, 1, 2) and Semantic Context (Homogeneous vs. Heterogeneous)

	Context				
	Heterogeneous		Homogeneous		
Number of intervening trials	*M*	*SD*	*M*	*SD*	Difference
0	564	71	602	67	38**
1	655	56	683	60	28*
2	619	61	648	49	30**

* $p < .05$. ** $p < .01$.

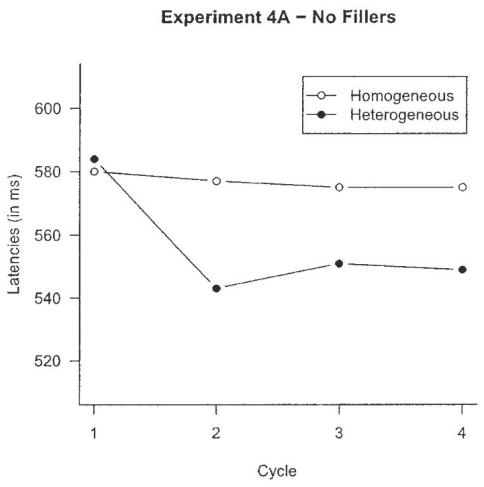
The refractory is preserved!!
It doesn't seem to be dramatically affected by the number of interleaved distractors.

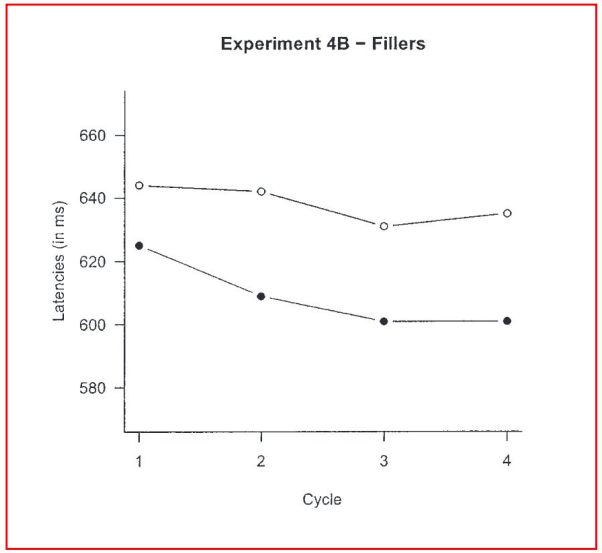
However: maybe people can keep the items' transient activation when no verbal distractor is presented

What happens if we interleave the related items with unrelated naming trials?

« cherry » « butterfly » « banana » « fork » « apple » « globe » « pear »

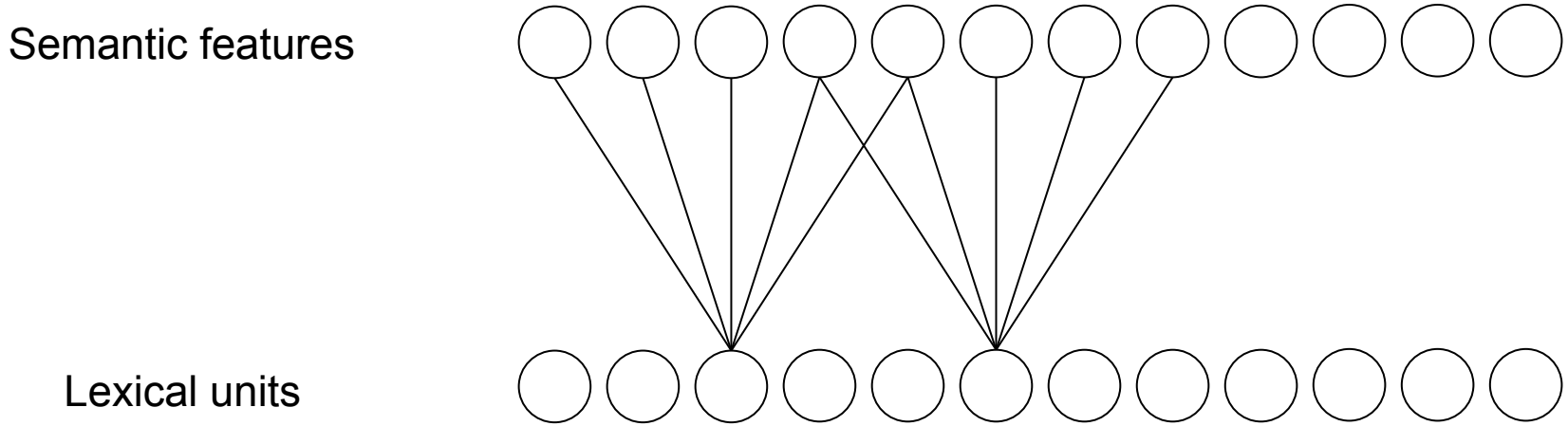
Time



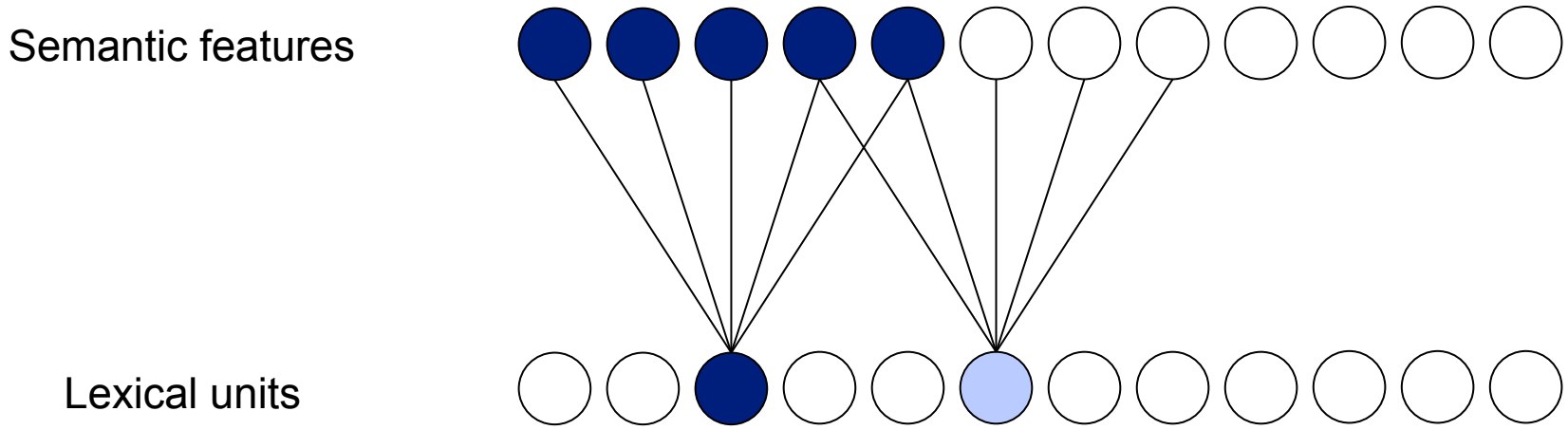


The blocking effect is preserved!

The fact that the blocking effect is still observed when homogeneous items are interleaved by heterogeneous items means that the effect cannot be attributed to residual semantic priming. This is because residual spreading of activation is supposed to be short-lived.

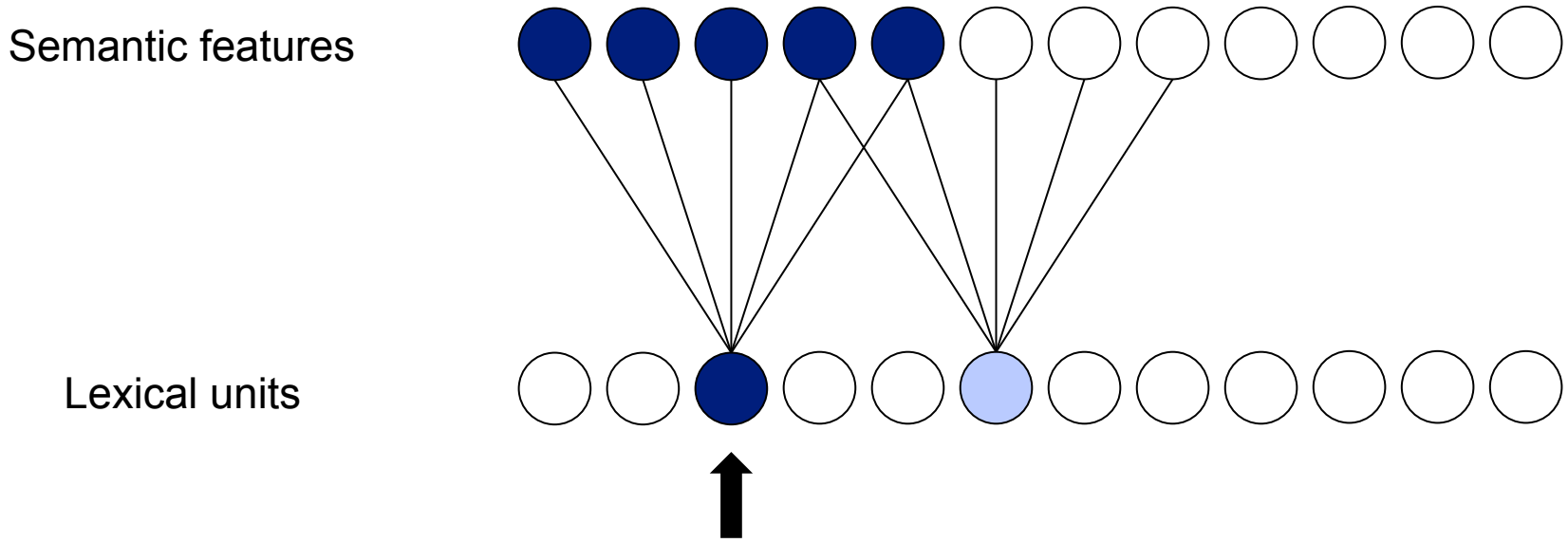


Oppenheim et al. (2010)



Competition based on activation level

Oppenheim et al. (2010)



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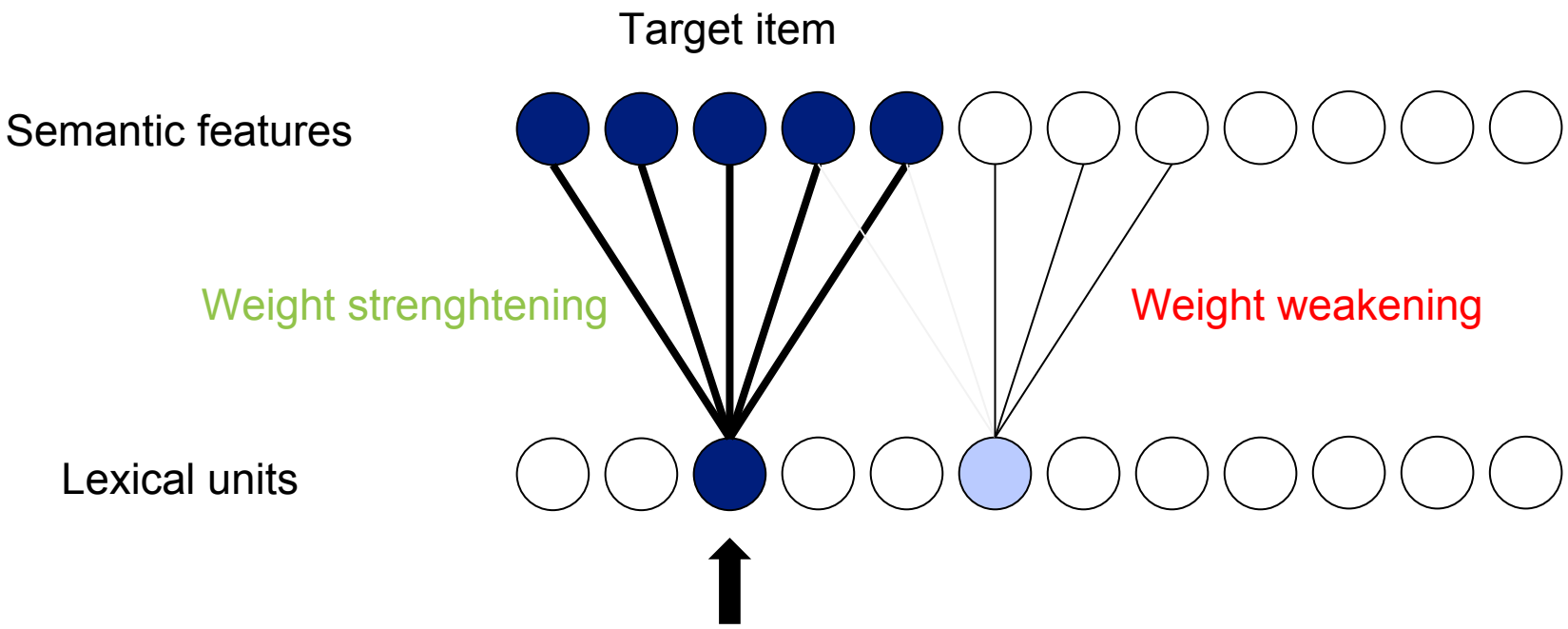
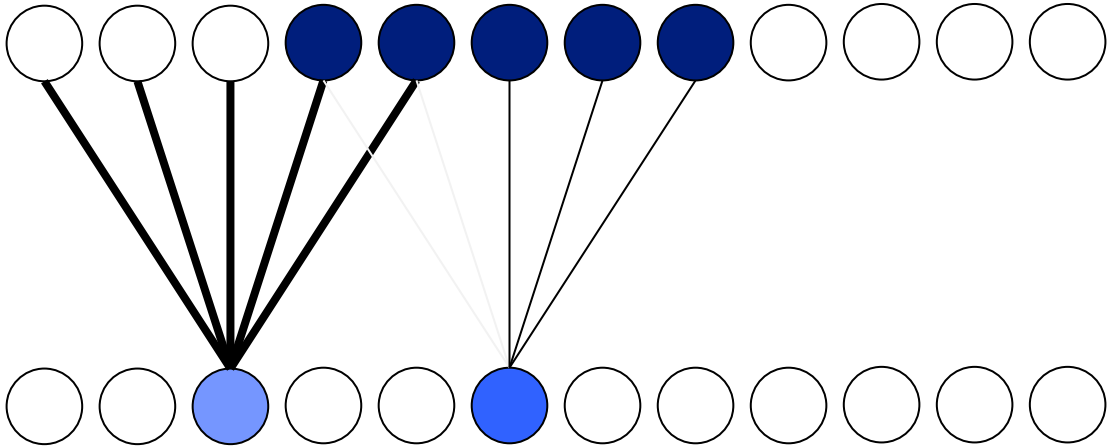


Oppenheim et al. (2010)

Semantically similar item

Semantic features



Lexical units

Increased competition!!

Oppenheim et al. (2010)

Response times increase within a cycle.
Explanation: increased competition (i.e., shared activation + incremental learning)

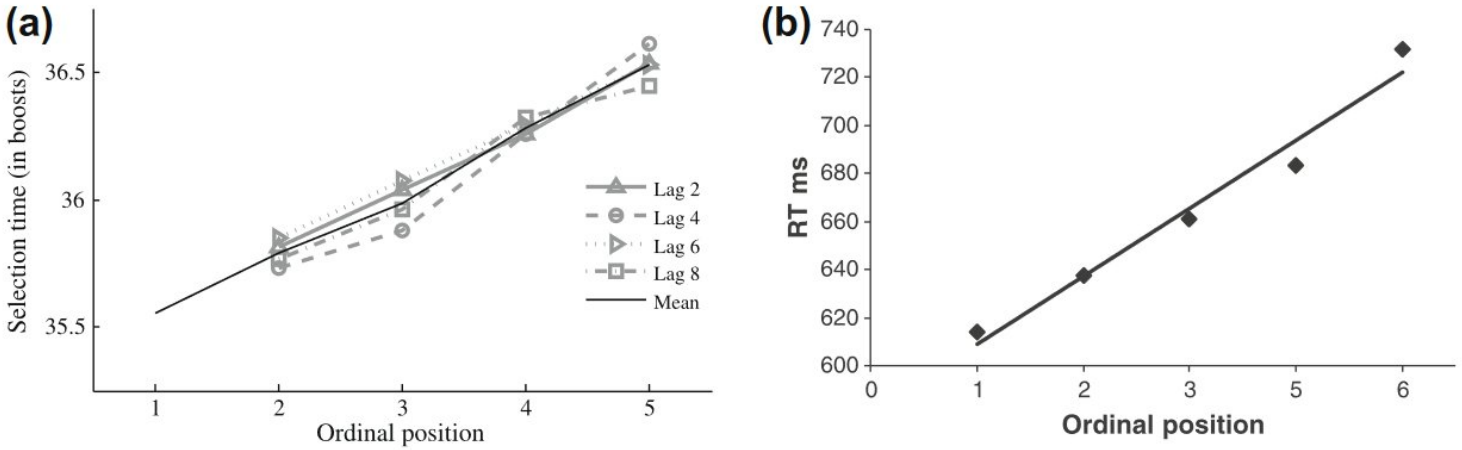


Fig. 3. In Simulation 1a, the continuous paradigm, each critical item takes a bit longer to select than the previous one, regardless of the number of intervening unrelated items. (a) Model-predicted mean correct selection times in each of the four lag conditions show identical increases with ordinal position. We present only an overall mean selection time for ordinal position 1 because the lag for the first item is obviously impossible to calculate. (b) Human subjects' mean naming latencies show the same increase. Reprinted from Howard et al., 2006.

Oppenheim et al. (2010)

Due to incremental learning (i.e., reinforcing the links between the target and its features), the model predicts repetition priming.

First item repeated

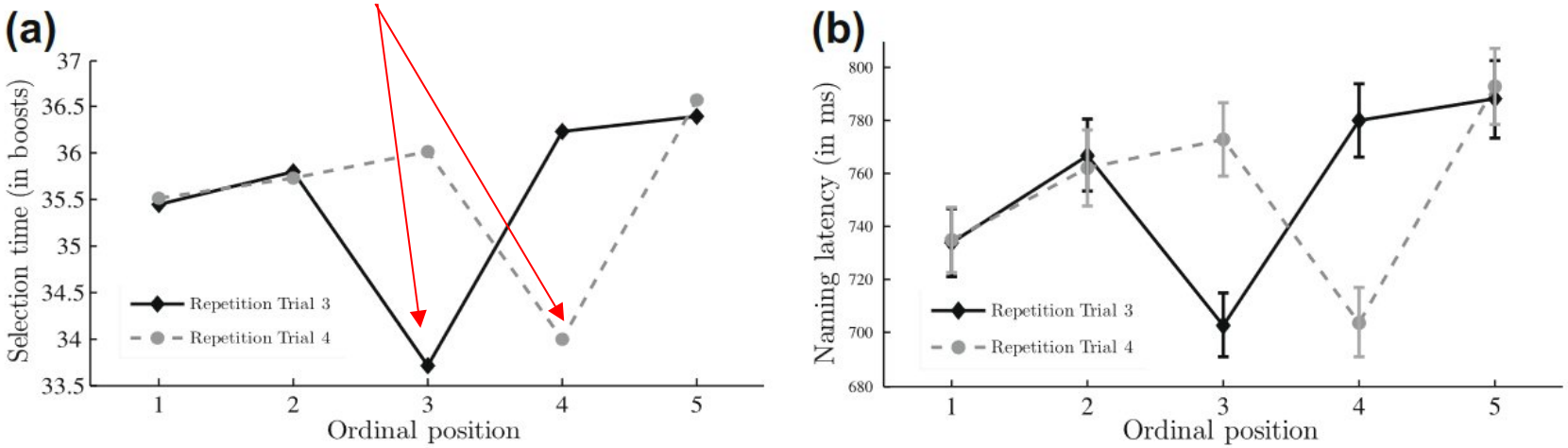


Fig. 4. Repeating an item creates cumulative semantic interference similar to that from accessing a novel item. (a) Model-predicted mean selection times from Simulation 1b. (b) Empirical results from Navarrete et al. (2008).

Oppenheim et al. (2010)

1. This decrease is due to general repetition priming

3. Bounce is due to a lack of repetition priming

4. Refractory effects generalizes to new items

2. The refractory effect appears progressively across cycles

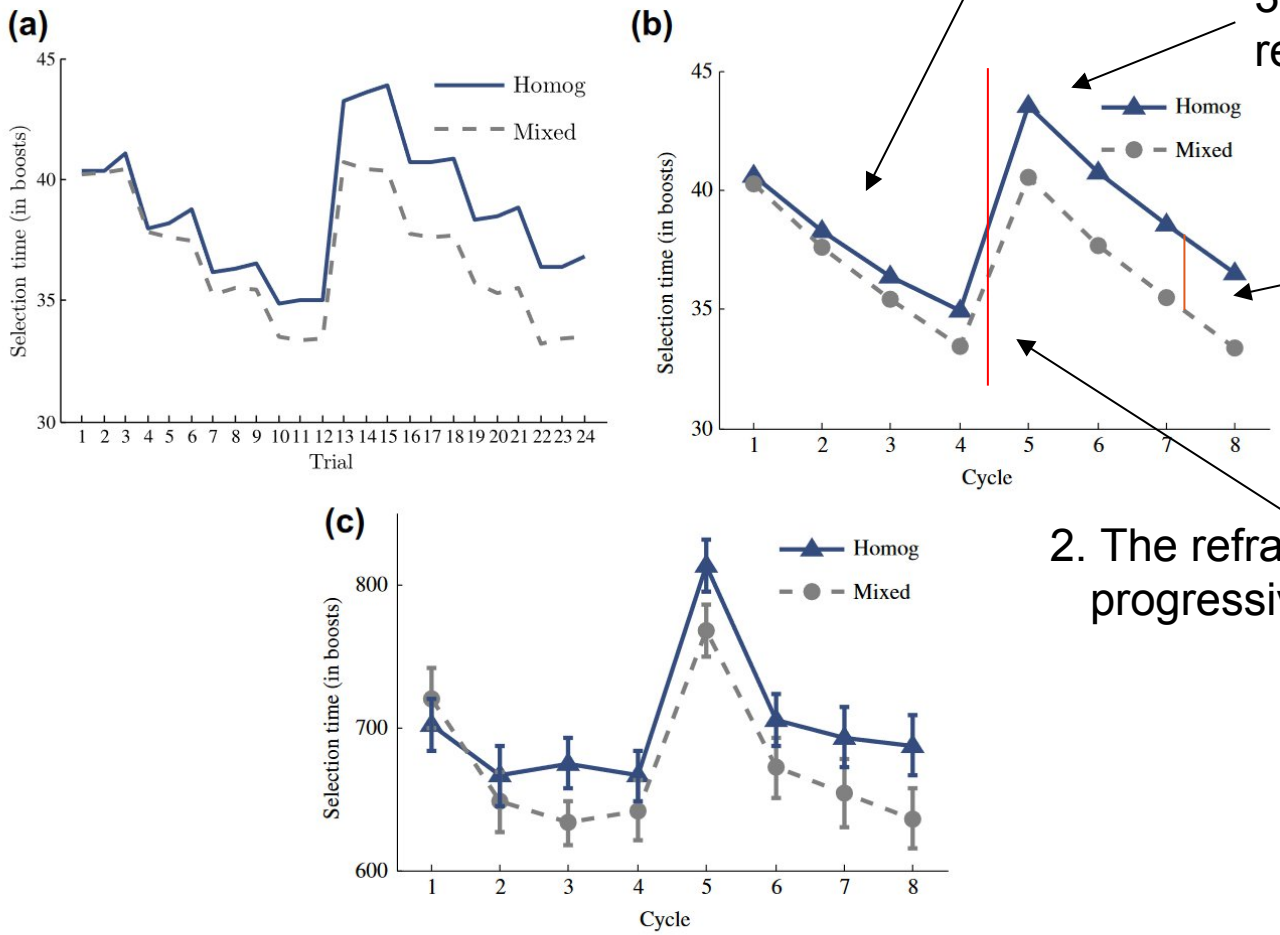


Fig. 6. Results from Simulation 3. (a) Lexical selection times for items in homogeneous and mixed sets presented in three-trial cycles. The second subset replaced the first after the twelfth trial. (b) Per-cycle mean selection latencies, derived from (a), for comparison with (c). (c) Human subjects' per-cycle mean naming latencies, from Belke et al. (2005).

Oppenheim et al. (2010)

Take-home message:

1. Naming latencies slow down when naming items from the same category
2. Due to a problem of selection at the semantic level
3. Is not due to short-lived semantic priming
4. The best explanation so far boils down to an incremental learning process

Thank you for you attention